tec-speed 20.0 VT-870 H1000

UL Approval: E214381 Version:20/11/2024

PROCESS GUIDE

Precautions in Handling

Storage Condition & Shelf Life

		Laminate
Storage Condition	Temperature	Room
	Relative Humidity	/
Shelf Life		24 Months(airproof)

Designing and Inner layer Process

- Dimension stability is superior to Standard FR-4 materials.
- Please consult with your oxide vendor to make sure that our material is fully compatible with your oxide process. We recommend controlling the peel strength to over 2 lb./in.
- After oxide, it is preferred to press cores within 12 hours. Any cores staged more than 24 hours should be re-treated.
- Please confirm the grinding treatment process as the material was soft and brittle. Please don't use the grinding process for the inner and outer layer with large area or full no copper design.

Registration system

- Strongly recommend Post Etch Punch and Pin technology as the basis for the inner layer registration systems.
- If customer utilizes fusion method, below is the requirement for fusing the material. Please check with the machine supplier, and test the fusion performance firstly.
- Operate temperature:≥200°C.
- Dwell time: At least 1min.

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PROCESS GUIDE

Availability-Laminate

Laminate	Core Thickness (inch)	Dk@10GHz	Df@10GHz
		IPC-TM650 2.5.5.5 Clamped Stripline	Cavity resonator
VT-870 H1000	0.025	10.2±0.25	0.003
	0.025	10.2±0.25	0.003
	0.050	10.2±0.25	0.003
	0.100	10.2±0.25	0.003

Remarks:

- Copper foil: Hoz~2oz, HTE, RTF and HVLP are all available based upon request.
- More types may be available upon request.

Bake Cycles during PCB Fabrication

There are a few optional baking cycles that may be used PCB fabrication. Please take into consideration when appropriate.

Process	Purpose	Cycle
After Cutting (for laminate <10 mil)	Dimensional stabilization; Eliminate moisture	2 hours150°C
After oxide	Eliminate moisture	1-2 hours 120°C
After Drilling	Eliminate stress	2 hours@ 190°C
Before HASL	Eliminate moisture	2 hours@ 125°C
Before shipment (if stored for over 3 months)	Eliminate moisture	2-4 hours@ 105°C

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PROCESS GUIDE

Typical Drilling Parameters

- Drilling parameters should be adjusted depending on hole size, layer count, panel thickness, stack count and stack height etc.
- Please adjust drilling parameters after checking qualities of through holes.

Suggest Drilling parameter as blow:

Diameter (mm)	Spindle speed (KRPM)	Feed rate (IPM)	Retract (IPM)	Hit Counts
0.20	55	75	500	500
0.25	55	75	500	500
0.30	40	52	500	500
0.40	55	75	1000	500
0.60	46	70	1000	500
1.00	38	76	1000	500
2.53	21	63	1000	500

- Excessive wear of carbide drill bits. Diamond coated drill bits are recommended.
- LE Aluminum sheet and white phenolic entry board are preferred, please consult your local Ventec representative for options.

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PROCESS GUIDE

Desmear Process

Panels should be baked for at least 1 hour at 120°C prior to plasma treatment. Plasma treated holes are more delicate than permanganate etched holes. Panels should not be exposed to any pressure wash or scrubbing process prior to metallization.

Plasma Desmear:
One time of FR-4 conditions is recommendable.

Permanganate horizontal Desmear:
The weight loss of VT-870 H1000 laminate is more than that of our conventional FR-4 material. One time of FR-4 condition is recommendable.

Permanganate vertical Desmear:
Vertical desmear process is not recommendable.

Packaging and baking recommendation

- It is recommended to bake the board before packaging at 125°C/4~8h to avoid moisture causing a decrease in heat resistance.
- If the PCBs needs to be stored for a long time before use, it is recommended to use aluminum foil vacuum packaging.
- If exceed 3 months after packaging , It is best to bake the PCBs at 125°C/4~6h before assembly before use.